

Challenge (Habanero)

- Q1.** What is the factored form of $x^2 - 16$?
(A) $(x + 4)(x - 4)$
(B) $x(x - 4)$
(C) $x(x + 4)$
(D) $x^2 - 4^2$
(E) $x^2 + 4^2$
- Q2.** A hiking trail is 3 miles long, and a park ranger needs to mark the trail with signs every x miles. If the ranger has 9 signs and the trail is marked with a difference of squares pattern, how far apart are the signs in miles, given $x^2 - 9 = 0$?
(A) 1 mile
(B) 3 miles
(C) 5 miles
(D) 7 miles
(E) 9 miles
- Q3.** A species of bird has a population that grows according to the equation $P = x^2 - 64$, where x is the number of years since the species was first observed. What is the factored form of this equation?
(A) $P = (x + 8)(x - 8)$
(B) $P = x(x - 8)$
(C) $P = x(x + 8)$
(D) $P = x^2 + 64$
(E) $P = x^2 - 8^2$
- Q4.** What is the factored form of $25x^2 - 49y^2$?
(A) $(5x + 7y)(5x - 7y)$
(B) $(5x - 7y)(5x + 7y)$
(C) $x(5x - 7y)$
(D) $y(5x + 7y)$
(E) $x(5x + 7y)$
- Q5.** A fuel tank can hold $x^2 - 36$ gallons of fuel, where x is the number of hours the fuel pump has been running. If the pump has been running for 6 hours, how many gallons of fuel are in the tank?
(A) 0 gallons
(B) 36 gallons
(C) -36 gallons
(D) 6 gallons
(E) 60 gallons
- Q6.** A mountain has a height of $x^2 - 49$ feet, where x is the number of miles from the base of the mountain. What is the factored form of this equation?
(A) $x(x - 7)$
(B) $x(x + 7)$
(C) $x^2 - 7^2$
(D) $x^2 + 7^2$
(E) $x^2 + 49$
- Q7.** A car's fuel consumption is given by the equation $C = x^2 - 9$, where x is the number of hours driven. If $x = 5$, what is the fuel consumption in gallons?
(A) 16 gallons
(B) 20 gallons
(C) 25 gallons
(D) 30 gallons
(E) 36 gallons
- Q8.** What is the factored form of $4x^2 - 9y^2$?
(A) $2x + 3y$
(B) $2x - 3y$
(C) $(2x + 3y)(2x - 3y)$
(D) $2(2x + 3y)$
(E) $3(2x - 3y)$

Open Ended Questions (FRQ)

- Q9.** A hiker is walking on a trail with an elevation change given by $h = x^2 - 25$. Find the elevation when $x = 6$. Show your work and explain your answer.
- Q10.** A park has a population of animals that grows according to $P = x^2 - 100$. What is the population when $x = 12$? Use the difference of squares to factor the equation and find the solution. Provide a brief explanation.

Chef's Tips

- Tip 1: Encourage students to check their work carefully.
- Tip 2: Remind students to apply the difference of squares formula correctly.
- Tip 3: Emphasize the importance of showing all work for free response questions.